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VIAL HOLDER

Abstract

Various exemplary methods, systems, and devices for vial holders are provided. The vial holders are assembled from housing components with a two-step snap closure, where a first step partially assembles the vial holders into a stable first configuration to form vial slots and shelf platforms for case of receiving vials. The vial holders are advantageous for holding and labeling vials containing temperature sensitive therapeutic formulations that need to be handled under conditions limited by time and dexterity by permitting partial pre-assembly and labeling at a remote location and facilitating final assembly with minimal effort. In some embodiments, the vial holders incorporate tamper resistant structures that physically shield the two-step snap closure from external access.

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Background/Summary

FIELD

[0001] The present disclosure relates generally to devices for holding vials, and in particular to vial holders comprising housings assembled with a two-step snap closure.

BACKGROUND

[0002] Certain therapeutic formulations are very temperature sensitive. These formulations are maintained in vials at very low temperatures, with processing steps such as labeling and sorting necessarily performed quickly by operators donning bulky insulated gear. Due to the constraints posed by tight timing and limited dexterity, vial handling may be facilitated by providing a holder that can be assembled and manipulated at low temperatures with minimal effort.

[0003] Accordingly, there is a need for an improved vial holder with easy assembly and improved handling.

SUMMARY

[0004] In general, methods, systems, and devices for vial holders are provided.

[0005] In one aspect, a vial holder is provided, comprising: a housing comprising at least one vial slot; and a shelf positioned inferior to each vial slot; wherein the housing is assembled from a first and an opposing second housing component, each comprising a portion of each vial slot and each shelf; wherein the housing components are attached to each other by a two-step snap closure engagement between at least one clip and at least one clip retainer having an outer first stop and an inner second stop; wherein the outer first stop of each clip retainer is configured to engage a clip of an opposing housing component in a partially assembled configuration, and wherein the inner second stop of each clip retainer is configured to engage a clip of an opposing housing component in a fully assembled configuration.

[0006] In one aspect, the partially assembled configuration partially forms each vial slot to receive a vial therein to rest upon each shelf.

[0007] In one aspect, the fully assembled configuration locks each received vial within each vial slot.

[0008] In one aspect, the at least one clip retainer and the at least one clip are positioned on an exterior of each housing component.

[0009] In one aspect, each clip abuts against each clip retainer at the outer first stop and is held in position by a friction fit.

[0010] In one aspect, each clip engages each clip retainer at the inner first stop.

[0011] In one aspect, each clip engages a first clip retainer at the outer first stop and engages a second clip retainer at the second inner stop.

[0012] In one aspect, the at least one clip retainer and the at least one clip are positioned on an interior of each housing component.

[0013] In one aspect, the at least one clip retainer is integrated into a first shelf portion and the at least one clip extends from a second shelf portion of each housing component.

[0014] In one aspect, the at least one clip retainer is positioned at two corners of each housing component and the at least one clip is positioned at opposing two corners of each housing component.

[0015] In one aspect, each of the first and second housing component comprises an inner edge that meets face to face in the fully assembled configuration to form a seam.

[0016] In one aspect, each of the first and second housing component comprises a plurality of fins positioned adjacent to and extending outwards past a segment of the inner edge.

[0017] In one aspect, the plurality of fins block the seam formed by the inner edges from exterior access in a fully assembled configuration.

[0018] In one aspect, the plurality of fins forms a tamper-resistant barrier to the two-step snap closure by blocking exterior access.

[0019] In one aspect, each vial slot is connected to at least one window.

[0020] In one aspect, an inferior face of the housing comprises one or more vents fluidly connected to an interior of the housing.

[0021] In one aspect, an inferior face of the housing is open.

[0022] In one aspect, an exterior surface of the housing is configured to receive one or more labels.

[0023] In one aspect, the first and second housing components are identical.

[0024] In one aspect, the first and second housing components are non-identical.

[0025] In one aspect, a method of assembling and loading a vial holder is provided, comprising the steps of: providing a first and second housing component, each housing component comprising a plurality of vial slot portion, shelf portions inferior to each vial slot portion, at least one clip, and at least one clip retainer having an outer detent and an inner detent; pressing the first and second housing components together such that each clip of each housing component seats within the outer detent of each clip retainer of the opposing housing component to form partially formed vial slots and partially formed shelves; positioning a vial in each of the partially formed vial slots to rest on the partially formed shelves; and pressing the first and second housing components together such that each clip of each housing component seats within the inner detent of each clip retainer of the opposing housing component to lock each vial in fully formed vial slots.

[0026] In one aspect, the providing step and first pressing step are performed at a location separate from the positioning step and second pressing step.

[0027] In one aspect, the method further comprises a step of applying a label to the first housing, the second housing, or both, prior to the positioning step.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The present disclosure is described by way of reference to the accompanying figures which are as follows:

[0029] FIG. 1 illustrates a side view of a fully assembled vial holder;

[0030] FIG. 2 is a perspective view illustrating unassembled vial holder housing components of FIG. 1;

[0031] FIG. 3 is a perspective view of a partially assembled vial holder of FIG. 1;

[0032] FIG. 4 is a perspective view of the placement of vials in the partially assembled vial holder of FIG. 3;

[0033] FIG. 5 is a perspective view of a fully assembled vial holder of FIG. 4;

[0034] FIG. 6 is a perspective view illustrating another unassembled vial holder housing component of FIG. 1;

[0035] FIG. 7 is a perspective view illustrating another vial holder housing component of FIG. 1;

[0036] FIG. 8 is a perspective view of unassembled vial holder housing components (bottom left) and a fully assembled vial holder with vial (top right);

[0037] FIG. 9 is a bottom view of a partially assembled (left) and fully assembled (right) vial holder of FIG. 8;

[0038] FIG. 10 is a perspective view of unassembled vial holder housing components (left) and a fully assembled vial holder with vial (right):

[0039] FIG. 11 is a perspective view of unassembled vial holder housing components (bottom left) and a fully assembled vial holder with vial (top right).

DETAILED DESCRIPTION

[0040] Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices, systems, and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. A person skilled in the art will understand that the devices, systems, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present disclosure is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present disclosure.

[0041] Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can be determined for any geometric shape. A person skilled in the art will appreciate that a dimension may not be a precise value but nevertheless be considered at about that value due to any number of factors such as manufacturing tolerances and sensitivity of measurement equipment. Sizes and shapes of the systems and devices, and the components thereof, can depend at least on the size and shape of components with which the systems and devices will be used.

[0042] Various exemplary methods, systems, and devices for vial holders are provided.

[0043] The various figures show embodiments of vial holders of the present disclosure. The devices and related methods are described herein in connection with vial holders that facilitate holder assembly, vial insertion, and labeling that would otherwise be difficult at low temperatures using bulky equipment. Advantageously, the vial holders can be labeled and partially assembled beforehand, such that low temperature operations can be limited to vial insertion and snap fitting the vial holder into a fully assembled configuration. Partial assembly is enabled by a two-step snap closure, wherein housing components clip together in a durable first configuration that forms vial slots sized to receive one or more vials. In some embodiments, the first configuration also forms shelf surfaces below the vial slots upon which the one or more received vials may stably sit without further operator input. Final assembly is achieved by pressing the housing components together from the first configuration into a second fully assembled configuration, whereby the one or more received vials are secured within the vial slots. In some embodiments, the vial holders are formed from identical halves such that complexity of manufacturing and assembly are reduced.

[0044] Referring now to the figures, FIG. 1 depicts an exemplary vial holder **100** according to one embodiment of the present disclosure. Vial holder **100** comprises a housing assembled from two housing components **102** each having a superior end **104** and an inferior end **106**. One or more vial slots **108** are positioned at superior end **104** wherein vials secured therein are accessible. One or more windows **110** are connected to each vial slot **108** permitting visual inspection of vials secured therein. In some embodiments, inferior end **106** comprises one or more vents **112**, permitting drainage of condensate. The exterior surface of each housing component **102** comprises ample label space **114**. While the depicted label space **114** is shown having a sticker label attached thereto, it should be understood that any desired label or marking may be applied, including but not limited to etched labels, printed labels, molded labels, written labels, and the like.

[0045] Referring now to FIG. 2, vial holder **100** is depicted in an unassembled configuration, revealing the interior of each housing component **102**. Each housing component **102** comprises a

portion of a shelf **116** positioned inferior to a portion of a vial slot **108** (here, each portion is depicted as half of a shelf **116** and vial slot **108**), such that a vial inserted into a vial slot **108** may rest upon a superior surface **118** of each shelf **116**. The two-step snap closure mechanism comprises clip retainers **117** sized to receive upper and lower clips **120**. In the embodiment depicted in FIG. 2, the hollow interior of shelf **116** forms clip retainer **117**, which comprises first, outer detents **122a** and second, inner detents **122b**. Each housing component **102** comprises a plurality of fins **124** positioned adjacent to and extending outwards past a segment of an inner edge **126**. In assembly, two housing components **102** are pressed together such that inner edges **126** of each housing component **102** are brought together to meet face to face to form a seam in vial holder **100**. The plurality of fins **124** are positioned in a complementary arrangement such that when two housing components **102** are pressed together, the plurality of fins **124** of each housing component **102** is configured to slide into an opposing housing component **102** such that most portions of the seam formed by the inner edges **126** comprise an adjacently positioned fin **124**. Accordingly, an assembled vial holder **100** comprises fins **124** blocking the seam formed by the inner edges **126** from exterior access, thereby forming a tamper-resistant barrier against access through the seam to the snap closure mechanism formed between clip retainers **117** and clips **120**.

[0046] Referring now to FIG. 3 through FIG. 5, the assembly of vial holder **100** using a first and second housing component **102** is now described. As recited elsewhere herein, each housing component **102** comprises clip retainers **117** sized to receive clips **120**. Accordingly, assembly is initiated by providing a first and second housing component **102** and aligning clips **120** of the first housing component **102** with clip retainers **117** of the second housing component **102**, which in turn aligns clips **120** of the second housing component **102** with clip retainers **117** of the first housing component **102**. The first and second housing components **102** are pressed towards each other such that clips **120** of each housing component **102** seats within first, outer detents **122a** of each clip retainer **117** (the first step of the two-step snap closure), thereby fixing vial holder **100** into a first configuration (FIG. 3). In some embodiments, each vial holder **100** can be assembled into the first configuration at room temperature or any other environment prior to introduction to a low temperature environment wherein vials are stored. In some embodiments, labels may be applied to housing components **102** before or after assembly into the first configuration. In the first configuration, vial slots **108** are partially formed and shelves **116** are partially joined together below each vial slot **108** to receive one or more vials in a stable manner (FIG. 4). The first and second housing components **102** are pressed towards each other in a final step such that clips **120** of each housing component **102** seats within second, inner detents **122b** of each clip retainer **117** (the second step of the two-step snap closure), closing vial holder **100** around each received vial in the fully assembled configuration (FIG. 5) with fully formed vial slots **108** and shelves **116**.

[0047] Referring now to FIG. 6, vial holder **200** is now described. Similar to vial holder **100**, vial holder **200** comprises a housing assembled from two housing components **202** with a two-step snap closure. Each housing component **202** comprises a portion of a shelf **216** positioned inferior to a portion of a vial slot **208**, such that a vial inserted into a vial slot **208** may rest upon a superior surface **218** of each shelf **216**. Inserted vials may be visually inspected through one or more windows **210** connected to each vial slot **208**. Vial holder **200** comprises a two-step snap closure mechanism between clip retainer **217** sized to receive a clip **220**. In the embodiment depicted in FIG. 6, clip retainer **217** is oriented vertically adjacent to vial slots **208**. Each clip retainer **217** forms a dual purpose of extending a clip **220** on one side and receiving a clip **220** on an opposing side, wherein a hollow interior of clip retainer **217** comprises a first detent **222a** and a second detent **222b**. First detent **222a** sits at an outer position of clip retainer **217** and second detent **222b** sits at an inner position of clip retainer **217**. Each housing component **202** comprises a plurality of vents **212** and a plurality of fins **224** positioned adjacent to and extending outwards past a segment of an inner edge **226**. In assembly, two housing components **202** are pressed together such that inner edges **226** of each housing component **202** are brought together to meet face to face to form a

seam in vial holder **200**. The plurality of fins **224** are positioned in a complementary arrangement such that when two housing components **202** are pressed together, the plurality of fins **224** of each housing component **202** is configured to slide into an opposing housing component **202** such that most portions of the seam formed by the inner edges **226** comprise an adjacently positioned fin **224**. Accordingly, an assembled vial holder **200** comprises fins **224** blocking the seam formed by the inner edges **226** from exterior access, thereby forming a tamper-resistant barrier against access through the seam to the snap closure mechanism formed between clip retainers **217** and clips **220**. Assembly of vial holder **200** using a first and second housing component **202** into a first configuration and a fully assembled configuration is performed similarly to vial holder **100** described elsewhere herein.

[0048] Referring now to FIG. 7, vial holder **300** is now described. Similar to vial holder **100**, vial holder **300** comprises a housing assembled from two housing components **302** with a two-step snap closure. Each housing component **302** comprises a portion of a shelf **316** positioned inferior to a portion of a vial slot **308**, such that a vial inserted into a vial slot **308** may rest upon a superior surface **318** of shelf **316**. Inserted vials may be visually inspected through one or more windows **310** connected to each vial slot **308**. Vial holder **300** comprises a two-step snap closure mechanism between clip retainer **317** sized to receive a clip **320**. In the embodiment depicted in FIG. 7, clip retainer **317** is positioned at two corners of housing component **302** with pin-like clips **320** positioned at the remaining two corners of housing component **302**. Each clip retainer **317** is sized to receive a clip **320** and comprises a hollow interior having a first detent **322a** and a second detent **322b**. First detent **322a** sits at an outer position of clip retainer **317** and second detent **322b** sits at an inner position of clip retainer **317**. Each housing component **302** comprises a plurality of vents **312** and a plurality of fins **324** positioned adjacent to and extending outwards past a segment of an inner edge **326**. In assembly, two housing components **302** are pressed together such that inner edges **326** of each housing component **302** are brought together to meet face to face to form a seam in vial holder **300**. The plurality of fins **324** are positioned in a complementary arrangement such that when two housing components **302** are pressed together, the plurality of fins **324** of each housing component **302** is configured to slide into an opposing housing component **302** such that most portions of the seam formed by the inner edges **326** comprise an adjacently positioned fin **324**. Accordingly, an assembled vial holder **300** comprises fins **324** blocking the seam formed by the inner edges **326** from exterior access, thereby forming a tamper-resistant barrier against access through the seam to the snap closure mechanism formed between clip retainers **317** and clips **320**. Assembly of vial holder **300** using a first and second housing component **302** into a first configuration and a fully assembled configuration is performed similarly to vial holder **100** described elsewhere herein.

[0049] In some embodiments, the present invention relates to vial holders having external two-step snap closures. Referring now to FIG. 8, an exemplary vial holder **400** is now described. Similar to vial holder **100**, vial holder **400** comprises a housing assembled with a two-step snap closure from two housing components **402** each having a superior end **404** and an inferior end **406**. While vial holder **400** is shown with an open inferior end, it should be understood that closed ends and vented ends are also contemplated. Each housing component **402** comprises a shelf **416** positioned inferior to a portion of a vial slot **408**, such that a vial inserted into a vial slot **408** may rest upon a superior surface **418** of each shelf **416**. Shelf **416** can interlock with a shelf **416** of an opposing housing component **402** to form a larger surface to receive a vial (FIG. 9). Inserted vials may be visually inspected through one or more windows **410** connected to each vial slot **408**. Vial holder **400** comprises a two-step snap closure mechanism between clips **420** and tabs **417**. In the embodiment depicted in FIG. 8, housing component **402** comprises two tabs **417** configured to engage a clip **420** at a first stop **422a** and a second stop **422b**, wherein first stop **422a** sits at an outer position and second stop **422b** sits at an inner position. Each housing component **402** comprises a plurality of fins **424** that overlap each other and form a flush surface with an adjacent clip **420** in assembly. In

some embodiments, vial holder **400** further comprises one or more interlocking pins **428** and pinholes **430** to enhance rigidity and facilitate assembly.

[0050] The assembly of vial holder **400** using a first and second housing component **402** is similar to the assembly of vial holder **100** described elsewhere herein. Assembly is initiated by aligning clips **420** of a first housing component **402** with tabs **417** of a second housing component **402**, which in turn aligns clips **420** of the second housing component **402** with tabs **417** of the first housing component **402**. If present, pins **428** of each housing component **402** are also aligned with corresponding pinholes **430** of each housing component **402**. The first and second housing components **402** are pressed towards each other such that clips **420** of each housing component **402** clip over an outer tab **417** to rest at first stop **422a** (the first step of the two-step snap closure), thereby fixing vial holder **400** into a first configuration (FIG. **9**, left). In some embodiments, each vial holder **400** can be assembled into the first configuration at room temperature or any other environment prior to introduction to a low temperature environment wherein vials are stored. In some embodiments, labels may be applied to housing components **402** before or after assembly into the first configuration (a physical marking **415** may be provided to aid in label placement). In the first configuration, vial slots **408** are partially formed and shelves **416** are partially interlocked below each vial slot **408** to receive one or more vials in a stable manner (FIG. **9**, left). The first and second housing components **402** are pressed towards each other in a final step such that clips **420** of each housing component **402** clips over an inner tab **417** to rest at second stop **422b** (the second step of the two-step snap closure), closing vial holder **400** around each received vial in the fully assembled configuration (FIG. **8**, top right; FIG. **9**, right).

[0051] Referring now to FIG. **10**, an exemplary vial holder **500** is now described. Similar to vial holder **400**, vial holder **500** comprises a housing assembled with a two-step snap closure from two housing components **502** each having a superior end **504** and an inferior end **506**. While vial holder **500** is shown with an open inferior end, it should be understood that closed ends and vented ends are also contemplated. Each housing component **502** comprises a shelf **516** positioned inferior to a portion of a vial slot **508**, such that a vial inserted into a vial slot **508** may rest upon a superior surface **518** of each shelf **516**. Shelf **516** can interlock with a shelf **516** of an opposing housing component **502** to form a larger surface to receive a vial. Inserted vials may be visually inspected through one or more windows **510** connected to each vial slot **508**. Vial holder **500** comprises a two-step snap closure mechanism between clips **520** and tab **517**. In the embodiment depicted in FIG. **10**, housing component **502** comprises a tab **517** configured to abut against a clip **520** at a first stop **522a** with a friction fit and to engage clip **520** at a second stop **522b**, wherein first stop **522a** sits at an outer position and second stop **522b** sits at an inner position. Each housing component **502** comprises a plurality of fins **524** that overlap each other and form a flush surface with an adjacent clip **520** in assembly. In some embodiments, vial holder **500** further comprises one or more interlocking pins **528** and pinholes **530** to enhance rigidity and facilitate assembly.

[0052] The assembly of vial holder **500** using a first and second housing component **502** is similar to the assembly of vial holder **400** described elsewhere herein. Assembly is initiated by aligning clips **520** of a first housing component **502** with tab **517** of a second housing component **502**, which in turn aligns clips **520** of the second housing component **502** with tab **517** of the first housing component **502**. If present, pins **528** of each housing component **502** are also aligned with corresponding pinholes **530** of each housing component **502**. The first and second housing components **502** are pressed towards each other such that clips **520** of each housing component **502** abut against tabs **517** to rest at first stop **522a** (the first step of the two-step snap closure), thereby holding vial holder **500** into a first configuration by a friction fit. In the first configuration, vial slots **508** are formed and shelves **516** are partially interlocked below each vial slot **508** to receive one or more vials in a stable manner. The first and second housing components **502** are pressed towards each other in a final step such that clips **520** of each housing component **502** clips over tab **517** to rest at second stop **522b** (the second step of the two-step snap closure), closing vial holder

500 around each received vial in the fully assembled configuration (FIG. 10, top right). [0053] Referring now to FIG. 11, an exemplary vial holder **600** is now described. Similar to vial holder **500**, vial holder **600** comprises a housing assembled with a two-step snap closure from two housing components **602** each having a superior end **604** and an inferior end **606**. While vial holder **600** is shown with an open inferior end, it should be understood that closed ends and vented ends are also contemplated. Each housing component **602** comprises a shelf **616** positioned inferior to a portion of a vial slot **608**, such that a vial inserted into a vial slot **608** may rest upon a superior surface **618** of each shelf **616**. Shelf **616** can interlock with a shelf **616** of an opposing housing component **602** to form a larger surface to receive a vial. Inserted vials may be visually inspected through one or more windows **610** connected to each vial slot **608**. Vial holder **600** comprises a two-step snap closure mechanism between clips **620** and tabs **617**. In the embodiment depicted in FIG. 11, housing component **602** comprises a tab **617** configured to abut against each clip **620** at a first stop **622a** with a friction fit and to engage each clip **620** at a second stop **622b**, wherein first stop **622a** sits at an outer position and second stop **622b** sits at an inner position. In some embodiments, vial holder **600** can further comprises one or more interlocking pins and pinholes to enhance rigidity and facilitate assembly. Assembly of vial holder **600** using a first and second housing component **602** into a first configuration and a fully assembled configuration is performed similarly to vial holder **500** described elsewhere herein.

[0054] In various embodiments, the vial holders of the present invention can be constructed from any material, including metals, polymeric materials, or combinations of metals or polymeric materials known to persons having skill in the art. Contemplated polymeric materials include but are not limited to polyethylenes, polyamides, polyurethanes, polytetrafluoroethylenes, polycarbonates, polyimides, and combinations thereof. The vial holders of the present invention can be made using any suitable method known in the art. Contemplated methods include but are not limited to milling, casting, and molding. In some embodiments, the methods can include 3D printing or other additive manufacturing techniques commonly used in the art.

[0055] The vial holders of the present invention are not limited to the layout and arrangements described with respect to the depicted embodiments. It should be understood that any housing component configured for assembly with a two-step snap closure is contemplated. Vial holders are not limited by the number of available vial slots, which can include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more. Vial slots can be arranged in any desired pattern, including but not limited to linear patterns and grid patterns. Vial holders can be assembled from identical housing components as well as non-identical housing components. Vial holders assembled from identical housing components benefit from reduced tooling costs and increased ease of manufacturing. Identical housing components are also simple to assemble as any two identical housing components may be used, leading to increased supply chain efficiencies.

[0056] The present disclosure has been described above by way of example only within the context of the overall disclosure provided herein. It will be appreciated that modifications within the spirit and scope of the claims may be made without departing from the overall scope of the present disclosure.

Claims

1. A vial holder, comprising: a housing comprising at least one vial slot; and a shelf positioned inferior to each vial slot; wherein the housing is assembled from a first and an opposing second housing component, each comprising a portion of each vial slot and each shelf; wherein the housing components are attached to each other by a two-step snap closure engagement between at least one clip and at least one clip retainer having an outer first stop and an inner second stop; wherein the outer first stop of each clip retainer is configured to engage a clip of an opposing housing component in a partially assembled configuration, and wherein the inner second stop of

each clip retainer is configured to engage a clip of an opposing housing component in a fully assembled configuration.

2. The vial holder of claim 1, wherein the partially assembled configuration partially forms each vial slot to receive a vial therein to rest upon each shelf.

3. The vial holder of claim 2, wherein the fully assembled configuration locks each received vial within each vial slot.

4. The vial holder of claim 1, wherein the at least one clip retainer and the at least one clip are positioned on an exterior of each housing component.

5. The vial holder of claim 4, wherein each clip abuts against each clip retainer at the outer first stop and is held in position by a friction fit.

6. The vial holder of claim 5, wherein each clip engages each clip retainer at the inner first stop.

7. The vial holder of claim 4, wherein each clip engages a first clip retainer at the outer first stop and engages a second clip retainer at the second inner stop.

8. The vial holder of claim 1, wherein the at least one clip retainer and the at least one clip are positioned on an interior of each housing component.

9. The vial holder of claim 8, wherein the at least one clip retainer is integrated into a first shelf portion and the at least one clip extends from a second shelf portion of each housing component.

10. The vial holder of claim 8, wherein the at least one clip retainer is positioned at two corners of each housing component and the at least one clip is positioned at opposing two corners of each housing component.

11. The vial holder of claim 1, wherein each of the first and second housing component comprises an inner edge that meets face to face in the fully assembled configuration to form a seam.

12. The vial holder of claim 11, wherein each of the first and second housing component comprises a plurality of fins positioned adjacent to and extending outwards past a segment of the inner edge.

13. The vial holder of claim 12, wherein the plurality of fins block the seam formed by the inner edges from exterior access in a fully assembled configuration.

14. The vial holder of claim 13, wherein the plurality of fins forms a tamper-resistant barrier to the two-step snap closure by blocking exterior access.

15. The vial holder of claim 1, wherein each vial slot is connected to at least one window.

16. The vial holder of claim 1, wherein an inferior face of the housing comprises one or more vents fluidly connected to an interior of the housing.

17. The vial holder of claim 1, wherein an inferior face of the housing is open.

18. The vial holder of claim 1, wherein an exterior surface of the housing is configured to receive one or more labels.

19. The vial holder of claim 1, wherein the first and second housing components are identical.

20. The vial holder of claim 1, wherein the first and second housing components are non-identical.

21. A method of assembling and loading a vial holder, comprising the steps of: providing a first and second housing component, each housing component comprising a plurality of vial slot portion, shelf portions inferior to each vial slot portion, at least one clip, and at least one clip retainer having an outer detent and an inner detent; pressing the first and second housing components together such that each clip of each housing component seats within the outer detent of each clip retainer of the opposing housing component to form partially formed vial slots and partially formed shelves; positioning a vial in each of the partially formed vial slots to rest on the partially formed shelves; and pressing the first and second housing components together such that each clip of each housing component seats within the inner detent of each clip retainer of the opposing housing component to lock each vial in fully formed vial slots.

22. The method of claim 20, wherein the providing step and first pressing step are performed at a location separate from the positioning step and second pressing step.

23. The method of claim 21, further comprising a step of applying a label to the first housing, the second housing, or both, prior to the positioning step.

